

NEBRASKA CITY, NE, USA

WMO: 725541

Lat: 40.606N

Lon: 95.864W

Elev: 1153

StdP: 14.09

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04993

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.0	4.0	-8.5	3.6	2.3	-4.8	4.4	6.6	30.7	28.4	27.2	28.6	8.4	330	0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	93.1	75.4	90.5	74.6	88.1	73.9	79.6	87.8	77.7	85.9	76.1	84.2	12.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.2	148.0	84.5	75.3	138.3	82.5	73.4	129.7	80.3	44.0	87.1	42.1	85.6	40.3	84.4	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.5	23.8	20.5	DB	-7.8	98.3	4.3	2.6	-10.9	100.2	-13.4	101.8	-15.9	103.2	-19.0	105.2	(4)
(5)			WB	-8.4	82.3	4.3	1.6	-11.5	83.5	-14.0	84.5	-16.4	85.4	-19.5	86.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.7	26.0	28.7	42.5	52.7	63.4	73.4	76.3	73.8	67.7	54.6	41.8	29.6	
	DBStd	20.16	11.96	12.11	11.94	9.89	8.77	5.78	5.64	5.32	7.71	9.37	10.26	10.83	
	HDD50	2706	745	600	295	86	7	0	0	0	0	58	280	635	
	HDD65	5655	1209	1017	701	381	137	6	2	4	60	343	696	1099	
	CDD50	3677	0	4	63	166	424	702	816	737	530	200	34	1	
	CDD65	1151	0	0	4	11	89	258	353	276	140	20	0	0	
	CDH74	10570	0	0	42	160	805	2412	3437	2281	1225	202	6	0	
	CDH80	3845	0	0	6	34	239	891	1395	811	419	50	0	0	
Wind	WSAvg	9.7	10.4	10.4	11.3	12.0	10.5	9.1	7.1	6.8	8.4	9.5	10.5	10.1	
Precipitation	PrecAvg	32.8	0.9	1.1	1.9	3.0	5.2	5.2	4.5	2.9	2.9	2.4	1.8	1.0	
	PrecMax	52.2	1.5	3.5	4.1	7.3	11.8	9.6	19.0	8.0	6.2	7.1	3.8	3.0	
	PrecMin	25.2	0.1	0.1	0.0	0.7	1.5	2.6	0.7	0.5	0.6	0.1	0.0	0.0	
	PrecStd	6.5	0.4	0.9	1.3	1.5	2.8	1.9	3.9	1.7	1.6	1.7	1.0	0.8	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.5	67.7	79.2	84.8	91.7	95.0	99.0	94.7	92.5	86.2	73.4	59.9	
		MCWB	44.7	50.2	61.6	64.6	68.8	74.8	74.9	78.1	71.2	68.1	55.7	49.1	
	2%	DB	52.1	58.7	73.0	78.9	85.9	91.3	93.7	90.9	88.8	79.4	67.7	53.8	
		MCWB	42.6	47.9	56.0	60.2	68.2	73.4	76.9	76.1	71.6	63.4	54.8	46.6	
	5%	DB	46.1	51.4	67.4	73.4	82.0	88.5	90.7	88.1	85.4	74.8	62.3	48.8	
		MCWB	39.6	43.3	54.6	57.8	65.4	73.3	76.5	75.7	70.5	61.6	52.3	42.8	
	10%	DB	41.4	45.4	61.3	69.5	78.8	85.6	87.7	84.4	81.2	70.3	57.2	44.6	
		MCWB	36.9	39.5	51.7	56.3	64.4	71.8	75.6	73.5	67.4	58.4	48.4	39.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.3	53.7	64.2	66.4	73.8	80.1	82.3	81.6	77.0	72.6	60.9	54.6
			MCDB	55.0	61.4	73.2	79.0	82.9	89.2	90.3	90.8	84.8	80.4	67.3	56.3
2%		WB	43.4	47.3	60.9	63.4	71.2	77.3	80.0	78.9	74.9	68.5	57.1	47.4	
		MCDB	50.5	59.6	68.3	73.7	80.9	86.4	88.4	86.6	81.8	75.7	63.2	51.6	
5%		WB	40.2	43.5	56.9	60.7	69.0	75.4	78.2	77.0	73.1	64.4	53.4	43.0	
		MCDB	46.1	50.9	64.1	70.3	78.1	84.2	86.4	84.7	80.1	70.5	61.0	48.8	
10%		WB	37.0	39.4	51.9	58.1	66.8	73.7	76.6	75.2	71.0	60.1	49.5	39.7	
		MCDB	41.6	44.8	62.5	66.8	75.3	82.3	84.8	82.5	78.6	67.8	57.8	44.4	
Mean Daily Temperature Range	5% DB	MDBR	17.0	17.5	20.1	20.9	19.6	18.9	18.5	18.4	20.9	20.6	19.4	16.2	
		MCDBR	24.3	27.6	29.0	27.5	23.8	22.0	21.8	21.9	24.1	26.3	26.8	23.4	
		MCWBR	16.4	16.9	15.4	14.2	10.6	9.6	9.5	10.3	9.7	13.4	15.1	16.4	
	5% WB	MCDBR	22.5	25.7	23.6	23.7	20.1	19.5	18.6	19.0	19.7	21.5	24.2	21.4	
		MCWBR	15.7	16.9	13.9	14.1	10.1	10.5	10.0	10.6	9.8	13.8	16.0	16.0	
Clear-Sky Solar Irradiance	taub	0.276	0.311	0.324	0.377	0.407	0.413	0.420	0.416	0.373	0.331	0.297	0.282		
	taud	2.487	2.399	2.423	2.294	2.278	2.308	2.282	2.308	2.382	2.485	2.510	2.475		
	Ebn at Noon	285	287	295	284	277	274	271	269	274	276	273	272		
	Edh at Noon	24	30	33	40	42	41	42	39	34	28	23	22		
All-Sky Solar Radiation	RadAvg	669	950	1240	1567	1851	2063	2062	1804	1490	1054	738	563		
	RadStd	51	61	111	99	151	154	140	94	99	118	56	56		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+2.16	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page